23.1.2 Exercises - Nuclear binding energy, fission and fusion

1. Phosphorus-30 ($^{30}_{15}P$) was the first artificial radioactive nuclide to be produced in a laboratory. This was achieved by bombarding aluminium-27 ($^{27}_{13}Al$) with α-particles.

A partial nuclear equation to represent this reaction is

$$^{27}_{13}Al + \alpha \rightarrow {}^{30}_{15}P + \Phi$$

a. State the full nuclear notation for

i. the α-particle, [1]

ii. the particle represented by the symbol Φ. [1]

b. Data for the rest masses of the particles in the reaction are given in Fig.

particle	mass/u
$^{27}_{13}Al$	26.98153
α	4.00260
$^{30}_{15}P$	29.97830
Φ	1.00867

Calculate, for this reaction,

i. the change in the total rest mass of the particles,

mass change = ... u [2]

ii. the energy, in joule, equivalent to the mass change calculated in (i).

energy = ... J [2]

c. With reference to your answer in **(b)(i)**, comment on the energy of the α-particle such that the reaction can take place.

.....

.....[2]

2. One nuclear reaction that can take place in a nuclear reactor may be represented, in part, by the equation

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{95}_{42}Mo + {}^{139}_{57}La + 2{}^{1}_{0}n + \ldots\ldots\ldots\ldots + \text{energy}$$

Data for a nucleus and some particles are given in Fig.

nucleus or particle	mass / u
$^{139}_{57}La$	138.955
$^{1}_{0}n$	1.00863
$^{1}_{1}p$	1.00728
$^{0}_{-1}e$	5.49×10^{-4}

a. Complete the nuclear reaction shown above. [1]

b.

i. Show that the energy equivalent to 1.00u is 934MeV. [3]

ii. Calculate the binding energy per nucleon, in MeV, of lanthanum-139 ($^{139}_{57}La$).

binding energy per nucleon = ... MeV [3]

c. State and explain whether the binding energy per nucleon of uranium-235 ($^{235}_{92}U$) will be greater, equal to or less than your answer in **(b)(ii)**.

.....

.....

.....[3]

3.

a. State what is meant by nuclear fusion and nuclear fission.

nuclear fusion: .....

nuclear fission: .....[3]

b. A nuclear reaction which may, in the future, be used for the generation of electrical energy is

$$^{2}_{1}H + ^{3}_{1}H \rightarrow ^{4}_{2}He + x.$$

i. Name the particle x.

.....[1]

ii. Data for the binding energy per nucleon E_B of some nuclei are given in Fig. [3]

		binding energy per nucleon $E_B / 10^{-13}$ J
deuterium	$^{2}_{1}H$	1.7813
tritium	$^{3}_{1}H$	4.5285
helium	$^{4}_{2}He$	11.3290

a) State the binding energy per nucleon of x.

binding energy per nucleon = ... J

b) Calculate the energy change that takes place in this reaction.

energy change = ... J

iii. Use your answer in **(ii) part 2** to determine the energy release when 2.0g of deuterium ($^{2}_{1}H$) reacts with 3.0g of tritium ($^{3}_{1}H$).

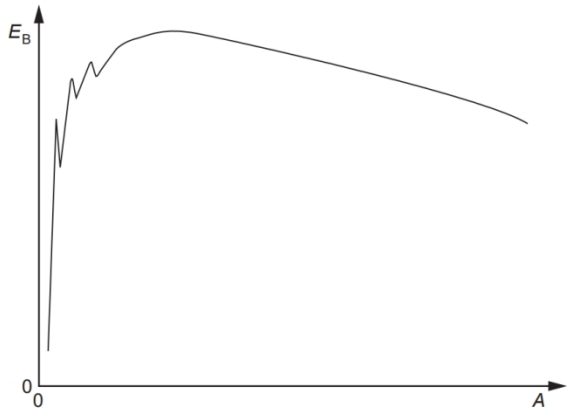
energy = ... J [1]

4.

a. Explain what is meant by the binding energy of a nucleus.

.....[1]

b. Fig. shows the variation with nucleon number (mass number) A of the binding energy per nucleon E_B of nuclei.



One particular fission reaction may be represented by the nuclear equation

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{141}_{56}Ba + ^{92}_{36}Kr + 3^{1}_{0}n.$$

i. On Fig, label the approximate positions of

a) the uranium ($^{235}_{92}U$) nucleus with the symbol U,

b) the barium ($^{141}_{56}Ba$) nucleus with the symbol Ba,

c) the krypton ($^{92}_{36}Kr$) nucleus with the symbol Kr. [2]

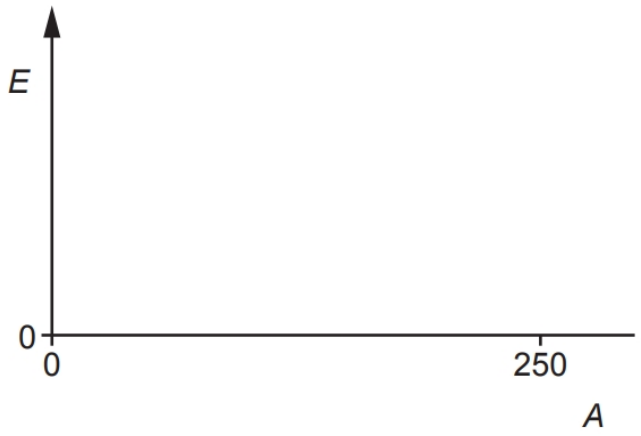
ii. The neutron that is absorbed by the uranium nucleus has very little kinetic energy. Explain why this fission reaction is energetically possible.

.....

.....[2]

5.

a. On Fig., sketch a line to show the variation with nucleon number A of the binding energy per nucleon E of a nucleus. [2]



b. In one type of nuclear process, deuterium ($^{2}_{1}H$) undergoes the reaction

$$^{2}_{1}H + ^{2}_{1}H \longrightarrow ^{3}_{2}He + ^{1}_{0}n.$$

i. State the name of this type of nuclear process.

.....[1]

ii. Explain, with reference to your line in **(a)**, why this reaction results in the release of energy.

.....

.....[2]

c. Table shows the masses of the particles involved in the reaction in (b).

particle	mass / u
$^{1}_{0}n$	1.008665
$^{2}_{1}H$	2.014102
$^{3}_{2}He$	3.016029

Calculate the energy released when 1.00mol of deuterium undergoes the reaction.

energy = ... J [5]